

UCH804: CHEMICAL PROCESS OPTIMIZATION AND STATISTICAL ANALYSIS

L	T	P	Cr
3	1	0	3.5

Course Objective:

To provide the skills for solving several types of optimization problems in process systems engineering.

Introduction: Process optimization, Formulation of various process optimization problems and their classification, Basic concepts of optimization-convex and concave functions, Necessary and sufficient conditions for stationary points.

Random Variables: Definition, Probability distribution, Distribution functions, Pdf and Cdf, Expectation and Variance.

Special Probability Distributions: Binomial, Poisson, Geometric, Uniform, Normal and Exponential distributions.

Sampling Distributions: Population and samples, Concept of sampling distributions, sampling distribution of mean, Chi-square, t and F distributions (pdf only).

Curve Fitting: Various methods of curve fitting, Regression and Correlation (Two variables only).

Optimization of single variable: Direct search methods, bracketing methods, Region elimination methods, Gradient-based methods, Root-finding using optimization techniques.

Optimization of multivariable: Optimality criteria, Unidirectional search, Direct search Methods. Gradient-based methods- Cauchy's (steepest descent) method, Newton's method.

Constrained Optimization Algorithms: Kuhn-Tucker conditions, Method of multipliers, Sensitivity analysis, Successive linear and quadratic programming, Optimization of staged and discrete processes.

Application of the above optimization techniques to chemical engineering processes: Reactors, heat exchanger design, separation processes, fluid flow systems.

Course Learning Outcomes (CLO):

Upon completion of the course, students will be able to:

1. apply probability and sampling distribution techniques to solve engineering problems.
2. solve problems related to curve fitting.
3. formulate the objective functions for constrained and unconstrained optimization problems.
4. use of different optimization techniques for chemical process design.

Text Books:

1. *Edgar, T. F., Himmelblau, D. M. and Lasdon, L.S. Optimization of Chemical Processes, McGraw-Hill (2001).*
2. *Babu, B.V., Process Plant Simulation, Oxford University Press (2004).*
3. *Gupta, S.C and Kapoor, V.K. Fundamentals of Mathematical Statistics, Sultan Chand & Sons(2002).*

Reference Books:

1. *Kalyanmoy, D., Optimization for Engineering Design, Prentice Hall (1998).*
2. *Reklaitis, G. V., Ravindran, A., and Ragsdell, K. M., Engineering Optimization - Methods and Applications, John Wiley (1983).*
3. *Box, G. E. P., Hunter, W. G., Hunter, J. S., Statistics for Experimenters – A n Introduction to Design, Data Analysis, and Model Building, John Wiley (1978).*
4. *Nguyen, H.T. and Roger, G.S., Fundamentals of mathematical statistics: Probability for statistics, Springer-Verlag New York (1989).*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessional (may include Quizzes/Assignments/Tutorials evaluations)	25